

Qatar-1b

R-0646

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230729 · created 2026-07-25T20:38:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460154.7919 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.016 +/- 0.054
Transit depth (Rp/Rs)^2	0.024 +/- 0.17 [%]
Orbital Inclination (inc)	80.12 +/- 2.97
Airmass coefficient 1 (a1)	0.42 +/- 0.24
Airmass coefficient 2 (a2)	0.66 +/- 1.65
Scatter in the residuals of the lightcurve fit is	58.05 %
Best Comparison Star	None
Optimal Aperture	15.25
Optimal Annulus	26.7
Transit Duration (day)	0.02 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.016 ± 0.054 vs 0.1463 (deviation 1.44σ)
Duration fit vs published	0.02 d vs 0.0692 d (deviation 1.18σ)
Mid-time O–C	1.8 min
Depth SNR	0.2
Residual scatter	58.05 %

Light curve

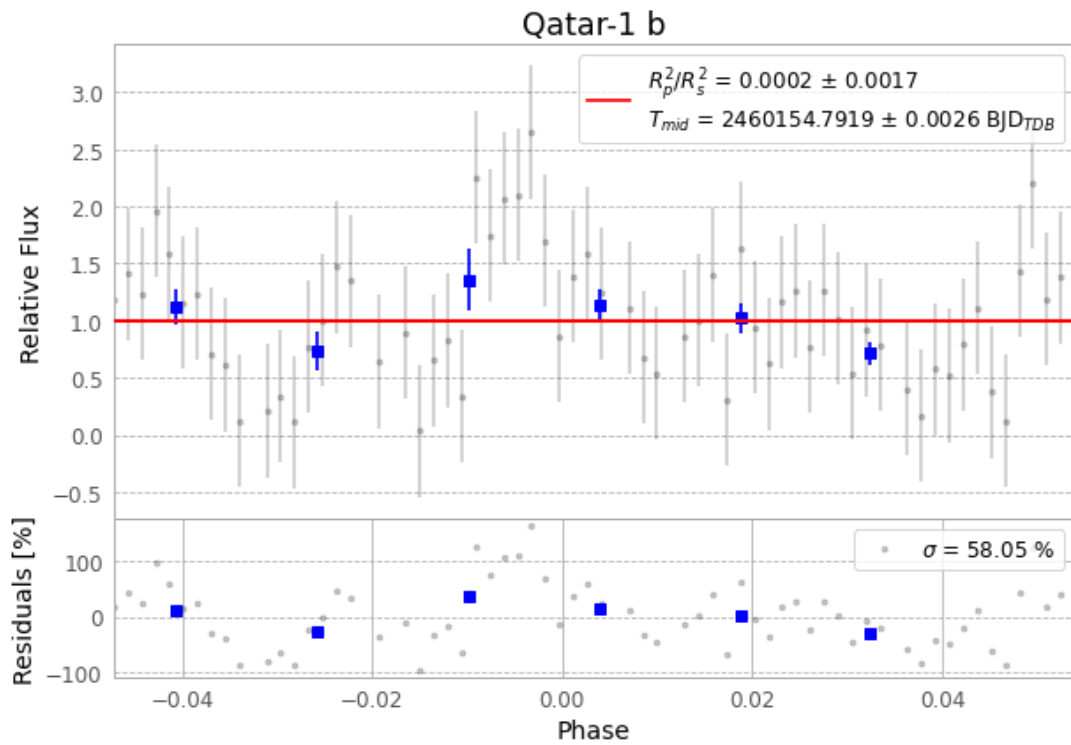


Figure 1 — FinalLightCurve_Qatar-1 b_2023-07-28.png (SHA-256 b6ec893061093d9e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [616, 311], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 845aee43f6f5655b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

70 FITS frames (46 MB), Qatar-1230729052115.FITS → Qatar-1230729084816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230729.log (9a90eb5068aa...), FinalLightCurve_Qatar-1 b_2023-07-28.png (b6ec89306109...), FinalLightCurve_Qatar-1 b_2023-07-28.csv (67d80f1b9d1f...)

Compute resources

Wall time 235.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 20:38Z.